

Saif Shikalgar

Robotics & Autonomous Systems Engineer

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Education

G. H. Raisoni College of Engineering and Management

Pune, India

B.Tech in Computer Science & Engineering (Cybersecurity Specialization); **CGPA: 8.10/10.0**

Sep 2023 – May 2027 (Expected)

Coursework: Microcontrollers & Embedded Systems, Control Systems, Operating Systems, Computer Networks, Object-Oriented Programming (OOP), Data Structures & Algorithms, DBMS.

Technical Skills

Robotics & Control Systems: Closed-Loop PID Control, Drive-by-Wire (DBW), Brake-by-Wire (BBW), Autonomous Emergency Braking (AEB), Adaptive Cruise Control (ACC), PWM/DAC Actuation, AS5600 12-Bit Encoders, Pneumatics.

Perception & Edge Computing: NVIDIA Jetson Nano/Orin, Sensor Fusion (Radar, IMU, Camera), YOLOv8 Object Detection, OpenCV, PyTorch, Real-Time Edge Inference, CUDA Acceleration.

Embedded Platforms & Protocols: STM32 (HAL/LL/CMSIS), ARM Cortex-M4, ESP32, FreeRTOS, CAN Bus 2.0B / TWAI, SPI, I2C, UART, BLE GATT, Microsecond-level Hardware Interrupts.

Languages & Simulation: C, C++ (C11/C++20), Python, MATLAB/Simulink, Docker, Linux (Ubuntu/Debian), Git.

Work Experience & Engineering Leadership

Team Abhyuday Racing

Pune, India

Technical Lead & Department Head – Drive-by-Wire (Robotics R&D)

Jan 2025 – Present

- Designed and deployed a distributed **3-ECU network (NVIDIA Jetson Nano + STM32F446RE + ESP32)** over CAN 2.0B bus with priority arbitration and 99.9% packet reliability under high EMI.
- Programmed real-time deterministic **Autonomous Emergency Braking (AEB)** control logic, securing **1st Place Nationally at aBAJA 2026** by halting safely at **6.2m** at 30 km/h with **40 bar hydraulic line pressure**.
- Implemented automotive-grade **failsafe supervision** with a **50 Hz CAN heartbeat loop**, 10 Hz telemetry scheduling, node-loss detection, and visual fault indicators, winning **1st Place in ACC** at aBAJA 2025.
- Engineered a modular Brake-by-Wire linear actuator system mounted parallel to the tandem master cylinder, cutting costs by **50%** and winning the **1st Place Manufacturing Excellence Award**.
- Developed MATLAB/Simulink vehicle kinematics models for autonomous trajectory execution, winning **1st Runner Up in MATLAB Advanced Simulation**.

AuraByte Studios (Independent Venture)

Pune, India

Lead Systems Developer & Founder

Dec 2025 – Present

- Built a graph-traversal algorithmic engine to compute complex fiber route loss models and sub-meter geospatial GPS plotting for 2.09K+ active users.

Key Robotics & Autonomous Projects

Dual-Stage Drive-by-Wire & Pneumatic Brake Actuation System | STM32F446RE, AS5600, C++, FreeRTOS, CAN Bus

- Engineered a closed-loop pneumatic brake control unit utilizing 12-bit magnetic angle encoders and fast-switching solenoid valves, achieving full 120 PSI brake pressure buildup in under 18ms.
- Implemented dual-channel cross-checking encoder logic to eliminate EMI-induced angle drift and guarantee fail-safe mechanical lockups.

Dual-Gate STM32 Hardware CAN Firewall & Bus Guardian | STM32F446RE, C11, CMSIS/HAL, FreeRTOS

- Created an inline deterministic CAN firewall intercepting malicious control signals with a **5.92 μ s processing latency**, preventing unauthorized throttle or steering overrides on robotic actuator buses.

24S LiFePO4 Battery Management System (BMS) Telematics Platform | ESP32, FreeRTOS, BLE GATT, Python, C++

- Reverse-engineered industrial JK BMS BLE GATT protocols to extract real-time cell voltage deltas (<15mV), current draws, and temperatures across a 76.8V 24S LiFePO4 battery pack with automated thermal runaway cutoff.

CYO: Edge AI Computer Vision Pipeline | PyTorch, YOLOv8, NVIDIA Jetson, CUDA, FastAPI

- Deployed YOLOv8 real-time object detection models with TensorRT optimization on edge hardware for dynamic obstacle identification and spatial bounding box estimation.

Honors & Awards

1st Place, Autonomous Emergency Braking (AEB): National aBAJA 2026 Competition; achieved a 6.2m halt on a 6.0m autonomous target threshold at 30 km/h.

1st Place, Manufacturing Excellence Award: National aBAJA 2026 Competition; recognized for modular Drive-by-Wire and Brake-by-Wire actuator design.

1st Runner Up, MATLAB Advanced Simulation: National aBAJA 2026 Competition; developed a GPS-based Point-A to Point-B autonomous traversal algorithm.

1st Place, Adaptive Cruise Control (ACC): aBAJA 2025; achieved following an 8-hour high-stakes crisis hardware migration from STM32 to ESP32.